

Exercise 1: Mocking and Stubbing

pom.xml: (adding dependencies)

```
<dependencies>
```

```
  <dependency>
```

```
    <groupId>org.junit.jupiter</groupId>
```

```
    <artifactId>junit-jupiter-api</artifactId>
```

```
    <version>5.10.0</version>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.junit.jupiter</groupId>
```

```
    <artifactId>junit-jupiter-engine</artifactId>
```

```
    <version>5.10.0</version>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.mockito</groupId>
```

```
    <artifactId>mockito-core</artifactId>
```

```
    <version>5.5.0</version>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.mockito</groupId>
```

```
<artifactId>mockito-junit-jupiter</artifactId>

<version>5.5.0</version>

<scope>test</scope>

</dependency>

</dependencies>
```

ExternalApi.java:

```
package com.example;

public interface ExternalApi {
    String getData();
}
```

MyService.java:

```
package com.example;

public class MyService {
    private ExternalApi externalApi;

    public MyService(ExternalApi externalApi) {
        this.externalApi = externalApi;
    }

    public String fetchData() {
        return externalApi.getData();
    }
}
```

MyServiceTest.java:

```
package com.example;

import static org.mockito.Mockito.*;
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;
import org.mockito.Mockito;
```

```

public class MyServiceTest {

    @Test
    public void testExternalApi() {

        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        when(mockApi.getData()).thenReturn("Mock Data");

        MyService service = new MyService(mockApi);

        String result = service.fetchData();
        assertEquals("Mock Data", result);
    }
}

```

Output:

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
[INFO] WARNING: A Java agent has been loaded dynamically (/Users/adityahargunani/.m2/repository/net/bytebuddy/byte-buddy-agent/1.14.6/byte-buddy-agent-1.14.6.jar)
[INFO] WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
[INFO] WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
[INFO] WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.399 s - in com.example.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.936 s
[INFO] Finished at: 2026-06-23T16:39:48+05:30
[INFO] -----
adityahargunani@Adityas-MacBook demo %

```